

Principles of Oral Surgery

OMFS 711

Dr Karim El Nokrashy DDS, MS

Learning Objectives & Competency Statements

Learning objectives

- **Perform a thorough pre-operative assessment including appropriate consultations, medical risk assessment, premedication, and incorporating various modifications to treatment. (competencies 2,3,4,5,6,8)**
- **Identify indications and contraindications and procedures for surgical extraction of erupted, partially, and fully impacted third molars. (competencies 2,3,4,5,6,8)**
- **Provide sound post-surgical management and effectively evaluate and treat post-operative complications (competencies 2,3,4,5,6,8)**

Course outline

- 1. Introduction/organization of course-Introduction of AAOMS Parameters of Care-Health Evaluation, Referral*
- 2. Principles of Surgery, Wound Repair*
- 3. Infection Control in Surgical Practice*
- 4. Pain and Anxiety Control in Surgical Practice, Hemostasis*
- 5. Instrumentation for Basic Oral Surgery, Principles of Routine Exodontia, Atraumatic Techniques*

- 6. Continue - Principles of Routine Exodontia, Atraumatic Techniques*
- 7. Principles of Complex Exodontia, Principles of Management of Impacted Teeth.*
- 8. Post Extraction Patient Management, Medicolegal considerations.*
- 9 Principles of Differential Diagnosis and Biopsy, Surgical management of Oral Pathologic Lesions*
- 10 Principles and Management of Odontogenic Infections, Odontogenic Diseases of the Maxillary Sinus*

11. Soft Tissue and Dentoalveolar Injuries.

Grading

In-Class Assessment/Quiz 20%

Course Project 30%

Objective Structured Clinical Exam (OSCE) 40%

Entrustable Professional Activities (EPA) 10%

Principles of Surgery

Surgery is a discipline based on principles that have evolved from basic research and centuries of trial and error. These principles pervade every area of surgery, whether oral and maxillofacial, periodontal, or gastrointestinal. This course provides information about patient health evaluation, managing medical emergencies, surgical concepts, principles of asepsis, and pain and anxiety control, which together form the necessary foundation for presentations of the specialized surgical techniques in succeeding lectures.

In this course we will discuss the process of evaluating the health status of patients. We will also describe methods of modifying surgical treatment plans to safely accommodate patients with the most common medical problems. Management of dental emergencies in the dental office will be briefly discussed as it was covered in introduction to oral Management of surgical complications and surgical emergencies will be covered in details.

Contemporary surgery is guided by a set of guiding principles, most of which apply no matter where in the body they are put into practice. Our 2nd presentation will cover the most important principles for those practitioners who perform surgery of the oral cavity and maxillofacial regions. Surgery always leaves a wound, whether one was initially present or not.

The surgical procedure is not complete when the final suture has been tied and the patient leaves. The surgeon's primary responsibility to the patient continues until the wound has healed, therefore an understanding of wound healing is mandatory for anyone who intends to create wounds surgically or manage accidental wounds.

In our third session we will discuss means of minimizing the risk of significant wound contamination and the spread of infectious organisms among individuals. This will include thorough decontamination of surgical instruments, disinfection of the room in which surgery is performed, lowering of bacterial counts in the operative site, and adherence to infection control principles by the members of the surgical team—in other words, strict adherence to aseptic technique.

In session 4 we will cover the common methods used by those performing oral surgery to control pain and anxiety, primarily through the use of local anesthesia and nitrous oxide sedation.

Then we will move on to present the armamentarium commonly used for exodontia. The basic instrumentation and the fundamental applications of instruments to their surgical purposes are illustrated and discussed. After that we will present the basic aspects of how to remove an erupted tooth atraumatically. The preoperative assessment and preparation of the patient will be discussed . The position of the patient in the chair and the position of the surgeon and the surgeon's hands for the removal of teeth in various parts of the mouth are detailed. The armamentarium and movements necessary to extract each type of tooth are presented in illustrations and descriptions.

We will then also discuss the basic aspects of managing complicated extractions (commonly termed surgical extractions). Surgical extractions primarily involve retrieving tooth roots and teeth that are likely to fracture, have fractured, or, for some other reason, pose an obstacle to extraction. In these situations, surgical removal of bone or surgical sectioning of the tooth is commonly required.

The fundamental aspects of the management of impacted teeth. The rationale for timely removal of impacted teeth is presented following surgical extractions. Classification and determination of the degree of difficulty of the impaction follows. Finally, a brief description of the basic surgical techniques required to remove impacted third molars will be provided.

A discussion of the management of the post-surgical patients and medicolegal consideration will follow that. Session nine will cover Principles of Differential Diagnosis and Biopsy, Surgical management of Oral Pathologic Lesions

Session ten we will discuss Principles and Management of Odontogenic Infections.

Our Last presentation will cover Soft Tissue and Dentoalveolar Injuries, for example alveolar fractures and soft tissue lacerations intra and extra orally.

Our last two sessions will have our project which will be a presentation on an assigned topic, that will count towards 30% of the course grade. Our finals will be an OSCE.

Preoperative Health Status Evaluation

The extent of the medical history, physical examination, and laboratory evaluation of patients requiring outpatient dentoalveolar surgery, under local anesthesia, nitrous oxide sedation, or both, differs substantially from that necessary for a patient requiring hospital admission and general anesthesia for surgical procedures. the dental professional must discover the presence or history of medical problems that may affect the safe delivery of the care she or he plans to provide, as well as any conditions specifically affecting the health of the oral and maxillofacial regions.

This is particularly true for patients for which dentoalveolar surgery is planned. This is due to several factors such as the added physical stress surgery patient experience, the creation of a bleeding wound that then needs to heal, the invasive nature of surgery that typically introduces microorganisms into the patient's tissues, and the common need for more elaborate pain and anxiety control measures and the use of more potent agents.

Standard Format for Recording Results of History and Physical Examinations

1. Biographic data
2. Chief complaint and its history
3. Medical history
4. Social and family medical histories
5. Review of systems
6. Physical examination
7. Laboratory and imaging results

Baseline Health History Database

1. Past hospitalizations, operations, traumatic injuries, and serious illnesses
2. Recent minor illnesses or symptoms
3. Medications currently or recently in use and allergies (particularly drug allergies)
4. Description of health-related habits or addictions, such as the use of ethanol, tobacco, and illicit drugs; and the amount and type of daily exercise
5. Date and result of last medical checkup or physician visit

Common Health Conditions to Inquire About Verbally or on a Health Questionnaire

- Allergies to antibiotics or local anesthetics
- Angina
- Anticoagulant use
- Asthma
- Bleeding disorders
- Breastfeeding
- Corticosteroid use
- Diabetes
- Heart murmurs
- Hepatitis
- Hypertension
- Implanted prosthetic devices
- Lung disease
- Myocardial infarction (i.e., heart attack)
- Osteoporosis
- Pregnancy
- Renal disease
- Rheumatic heart disease
- Seizure disorder
- Sexually transmitted diseases
- Tuberculosis

Routine Review of Head, Neck, and Maxillofacial Regions

- *Constitutional*: Fever, chills, sweats, weight loss, fatigue, malaise, loss of appetite
- *Head*: Headache, dizziness, fainting, insomnia
- *Ears*: Decreased hearing, tinnitus (ringing), pain
- *Eyes*: Blurring, double vision, excessive tearing, dryness, pain
- *Nose and sinuses*: Rhinorrhea, epistaxis, problems breathing through nose, pain, change in sense of smell
- *Temporomandibular joint area*: Pain, noise, limited jaw motion, locking
- *Oral*: Dental pain or sensitivity, lip or mucosal sores, problems chewing, problems speaking, bad breath, loose restorations, sore throat, loud snoring
- *Neck*: Difficulty swallowing, change in voice, pain, stiffness





Physical Examination Before Oral and Maxillofacial Surgery

Inspection

- *Head and face*: General shape, symmetry, hair distribution
- *Ear*: Normal reaction to sounds (otoscopic examination if indicated)
- *Eye*: Symmetry, size, reactivity of pupil, color of sclera and conjunctiva, movement, test of vision
- *Nose*: Septum, mucosa, patency
- *Mouth*: Teeth, mucosa, pharynx, lips, tonsils
- *Neck*: Size of thyroid gland, jugular venous distention

Palpation

- *Temporomandibular joint*: Crepitus, tenderness
- *Paranasal*: Pain over sinuses
- *Mouth*: Salivary glands, floor of mouth, lips, muscles of mastication
- *Neck*: Thyroid gland size, lymph nodes

Percussion

- *Paranasal*: Resonance over sinuses (difficult to assess)
- *Mouth*: Teeth

Auscultation

- *Temporomandibular joint*: Clicks, crepitus
- *Neck*: Carotid bruits

Brief Maxillofacial Examination

While interviewing the patient, the dentist should visually examine the patient for general shape and symmetry of head and facial skeleton, eye movement, color of conjunctiva and sclera, and ability to hear. The clinician should listen for speech problems, temporomandibular joint sounds, and breathing ability.

Routine Examination

Temporomandibular Joint Region

- Palpate and auscultate joints.
- Measure range of motion of jaw and opening pattern.

Nose and Paranasal Region

- Occlude nares individually to check for patency.
- Inspect anterior nasal mucosa.

Mouth

- Take out all removable prostheses.
- Inspect oral cavity for dental, oral, and pharyngeal mucosal lesions. Look at tonsils and uvula.
- Hold tongue out of mouth with dry gauze while inspecting lateral borders.
- Palpate tongue, lips, floor of mouth, and salivary glands (check for saliva).
- Palpate neck for lymph nodes and thyroid gland size. Inspect jugular veins.

American Society of Anesthesiologists (ASA) Classification of Physical Status

ASA I: A normal, healthy patient

ASA II: A patient with mild systemic disease or significant health risk factor

ASA III: A patient with severe systemic disease that is not incapacitating

ASA IV: A patient with severe systemic disease that is a constant threat to life

ASA V: A moribund patient who is not expected to survive without the operation

ASA VI: A declared brain-dead patient whose organs are being removed for donation purposes

General Anxiety-Reduction Protocol

Before Appointment

- Hypnotic agent to promote sleep on night before surgery (optional)
- Sedative agent to decrease anxiety on morning of surgery (optional)
- Morning appointment and schedule so that reception room time is minimized

During Appointment

Nonpharmacologic Means of Anxiety Control

- Frequent verbal reassurances
- Distracting conversation
- No surprises (clinician warns patient before doing anything that could cause anxiety)
- No unnecessary noise
- Surgical instruments out of patient's sight
- Relaxing background music

Pharmacologic Means of Anxiety Control

- Local anesthetics of sufficient intensity and duration
- Nitrous oxide
- Intravenous anxiolytics

After Surgery

- Succinct instructions for postoperative care
- Patient information on expected postsurgical sequelae (e.g., swelling or minor oozing of blood)
- Further reassurance
- Effective analgesics
- Patient information on who can be contacted if any problems arise
- Telephone call to patient at home during evening after surgery to check whether any problems exist

Box 1.10

Management of Patient With History of Angina Pectoris

1. Consult the patient's physician.
2. Use an anxiety-reduction protocol.
3. Have nitroglycerin tablets or spray readily available. Use nitroglycerin premedication, if indicated.
4. Ensure profound local anesthesia before starting surgery.
5. Consider the use of nitrous oxide sedation.
6. Monitor vital signs closely.
7. Consider possible limitation of amount of epinephrine used (0.04?mg maximum).
8. Maintain verbal contact with patient throughout the procedure to monitor status.

Box 1.11

Management of Patient With a History of Myocardial Infarction

1. Consult the patient's primary care physician.
2. Check with the physician if invasive dental care is needed before 6 months since the myocardial infarction.
3. Check whether the patient is using anticoagulants (including aspirin).
4. Use an anxiety-reduction protocol.
5. Have nitroglycerin available; use it prophylactically if the physician advises.
6. Administer supplemental oxygen (optional).
7. Provide profound local anesthesia.
8. Consider nitrous oxide administration.
9. Monitor vital signs, and maintain verbal contact with the patient.
10. Consider possible limitation of epinephrine use to 0.04?mg.
11. Consider referral to an oral-maxillofacial surgeon.

Box 1.12

Management of the Patient With Congestive Heart Failure (Hypertrophic Cardiomyopathy)

1. Defer treatment until heart function has been medically improved and the patient's physician believes treatment is possible.
2. Use an anxiety-reduction protocol.
3. Consider possible administration of supplemental oxygen.
4. Avoid using the supine position.
5. Consider referral to an oral-maxillofacial surgeon.

Box 1.13

Management of the Patient With Asthma

1. Defer dental treatment until the asthma is well controlled and the patient has no signs of a respiratory tract infection.
2. Listen to the chest with a stethoscope to detect any wheezing before major oral surgical procedures or sedation.
3. Use an anxiety-reduction protocol, including nitrous oxide, but avoid the use of respiratory depressants.
4. Consult the patient's physician about possible preoperative use of cromolyn sodium.
5. If the patient is or has been chronically taking corticosteroids, provide prophylaxis for adrenal insufficiency.
6. Keep a bronchodilator-containing inhaler easily accessible.
7. Avoid the use of nonsteroidal antiinflammatory drugs in susceptible patients.

Box 1.14

Management of Patient With Chronic Obstructive Pulmonary Disease

1. Defer treatment until lung function has improved and treatment is possible.
2. Listen to the chest bilaterally with stethoscope to determine adequacy of breath sounds.
3. Use an anxiety-reduction protocol, but avoid the use of respiratory depressants.
4. If the patient requires chronic oxygen supplementation, continue at the prescribed flow rate. If the patient does not require supplemental oxygen therapy, consult his or her physician before administering oxygen.
5. If the patient chronically receives corticosteroid therapy, manage the patient for adrenal insufficiency.
6. Avoid placing the patient in the supine position until you are confident that the patient can tolerate it.
7. Keep a bronchodilator-containing inhaler accessible.
8. Closely monitor respiratory rate and heart rate.
9. Schedule afternoon appointments to allow for clearance of secretions.

Box 1.15

Management of Patient With Renal Insufficiency and Patient Receiving Hemodialysis

1. Avoid the use of drugs that depend on renal metabolism or excretion. Modify the dose if such drugs are necessary. Do not use an atrioventricular shunt for giving drugs or for taking blood specimens.
2. Avoid the use of nephrotoxic drugs such as nonsteroidal antiinflammatory drugs.
3. Defer dental care until the day after dialysis has been given.
4. Consult the patient's physician about the use of prophylactic antibiotics.
5. Monitor blood pressure and heart rate.
6. Look for signs of secondary hyperparathyroidism.
7. Consider screening for hepatitis B virus before dental treatment. Take the necessary precautions if unable to screen for hepatitis.

Box 1.16

Management of Patient With Renal Transplant

1. Defer treatment until the patient's primary care physician or transplant surgeon clears the patient for dental care.
2. Avoid the use of nephrotoxic drugs. ^a
3. Consider the use of supplemental corticosteroids.
4. Monitor blood pressure.
5. Consider screening for hepatitis B virus before dental care. Take necessary precautions if unable to screen for hepatitis.
6. Watch for presence of cyclosporine-A-induced gingival hyperplasia. Emphasize the importance of oral hygiene.
7. Consider use of prophylactic antibiotics, particularly in patients taking immunosuppressive agents.

Most of these recommendations also apply to patients with other transplanted organs.

Box 1.17

Management of Patient With Hypertension

Mild to Moderate Hypertension (Systolic >140 mmHg; Diastolic >90 mmHg)

1. Recommend that the patient seek the primary care physician's guidance for medical therapy of hypertension. It is not necessary to defer needed dental care.
2. Monitor the patient's blood pressure at each visit and whenever administration of epinephrine-containing local anesthetic surpasses 0.04 mg during a single visit.
3. Use an anxiety-reduction protocol.
4. Avoid rapid posture changes in patients taking drugs that cause vasodilation.
5. Avoid administration of sodium-containing intravenous solutions.

Severe Hypertension (Systolic >200 mmHg; Diastolic >110 mmHg)

1. Defer elective dental treatment until the hypertension is better controlled.
2. Consider referral to an oral-maxillofacial surgeon for emergent problems.

Box 1.18

Management of Patient With Hepatic Insufficiency

1. Attempt to learn the cause of the liver problem; if the cause is hepatitis B, take usual precautions.
2. Avoid drugs requiring hepatic metabolism or excretion; if their use is necessary, modify the dose.
3. Screen patients with severe liver disease for bleeding disorders by using tests for determining platelet count, prothrombin time, partial thromboplastin time, and bleeding time.
4. Attempt to avoid situations in which the patient might swallow large amounts of blood.

Box 1.19

Management of Patient With Diabetes

Insulin-Dependent (Type 1) Diabetes

1. Defer surgery until the diabetes is well controlled; consult the patient's physician.
2. Schedule an early-morning appointment; avoid lengthy appointments.
3. Use an anxiety-reduction protocol, but avoid deep sedation techniques in outpatients.
4. Monitor pulse, respiration, and blood pressure before, during, and after surgery.
5. Maintain verbal contact with the patient during surgery.
6. If the patient must not eat or drink before oral surgery and will have difficulty eating after surgery, instruct him or her not to take the usual dose of regular or NPH insulin; start intravenous administration of a 5% dextrose in water drip at 150 mL/h.
7. If allowed, have the patient eat a normal breakfast before surgery and take the usual dose of regular insulin but only half the dose of NPH insulin.
8. Advise patients not to resume normal insulin doses until they are able to return to usual level of caloric intake and activity level.
9. Consult the physician if any questions concerning modification of the insulin regimen arise.
10. Watch for signs of hypoglycemia.
11. Treat infections aggressively.

Non-Insulin-Dependent (Type 2) Diabetes

1. Defer surgery until the diabetes is well controlled.
2. Schedule an early-morning appointment; avoid lengthy appointments.
3. Use an anxiety-reduction protocol.
4. Monitor pulse, respiration, and blood pressure before, during, and after surgery.
5. Maintain verbal contact with the patient during surgery.
6. If the patient must not eat or drink before oral surgery and will have difficulty eating after surgery, instruct him or her to skip any oral hypoglycemic medications that day.
7. If the patient can eat before and after surgery, instruct him or her to eat a normal breakfast and to take the usual dose of hypoglycemic agent.
8. Watch for signs of hypoglycemia.
9. Treat infections aggressively.

NPH, Neutral protamine Hagedorn.

Box 1.20

Management of Patient With Adrenal Suppression Who Requires Major Oral Surgery

If the patient is currently taking corticosteroids:

1. Use an anxiety-reduction protocol.
2. Monitor pulse and blood pressure before, during, and after surgery.
3. Instruct the patient to double the usual daily dose on the day before, day of, and day after surgery.
4. On the second postsurgical day, advise the patient to return to a usual steroid dose.

If the patient is not currently taking steroids but has received at least 20?mg of hydrocortisone (cortisol or equivalent) for more than 2 weeks within the past year:

1. Use an anxiety-reduction protocol.
2. Monitor pulse and blood pressure before, during, and after surgery.
3. Instruct the patient to take 60?mg of hydrocortisone (or equivalent) the day before and the morning of surgery (or the dentist should administer 60?mg of hydrocortisone or equivalent intramuscularly or intravenously before complex surgery).
4. On the first 2 postsurgical days, the dose should be dropped to 40?mg and dropped to 20?mg for 3 days thereafter. The clinician can cease administration of supplemental steroids 6 days after surgery.

If a major surgical procedure is planned, the clinician should strongly consider hospitalizing the patient. The clinician should consult the patient's physician if any questions arise concerning the need for or the dose of supplemental corticosteroids.

Box 1.21

Management of Patient With Hyperthyroidism

1. Defer surgery until the thyroid gland dysfunction is well controlled.
2. Monitor pulse and blood pressure before, during, and after surgery.
3. Limit the amount of epinephrine used.

Box 1.22

Management of Patient With a Coagulopathy

1. Defer surgery until a hematologist is consulted about the patient's management.
2. Have baseline coagulation tests, as indicated (prothrombin time, partial thromboplastin time, bleeding time, platelet count), and screening for hepatitis performed.
3. Schedule the surgery in a manner that allows it to be performed soon after any coagulation-correcting measures have been taken (after platelet transfusion, factor replacement, or aminocaproic acid administration).
4. Augment clotting during surgery with the use of topical coagulation-promoting substances, sutures, and well-placed pressure packs.
5. Monitor the wound for 2?h to ensure that a good initial clot forms.
6. Instruct the patient on ways to prevent dislodgment of the clot and on what to do should bleeding restart.
7. Avoid prescribing nonsteroidal antiinflammatory drugs.
8. Take precautions against contracting hepatitis during surgery.

Patients with severe coagulopathies who require major surgery should be hospitalized.

Box 1.23

Management of Patient Whose Blood Is Therapeutically Anticoagulated

Patients Receiving Aspirin or Other Platelet-Inhibiting Drugs

1. Consult the patient's physician to determine the safety of stopping the anticoagulant drug for several days.
2. Defer surgery until the platelet-inhibiting drugs have been stopped for 5 days.
3. Take extra measures during and after surgery to help promote clot formation and retention.
4. Restart drug therapy on the day after surgery if no bleeding is present.

Patients Receiving Warfarin (Coumadin)

1. Consult the patient's physician to determine the safety of allowing the prothrombin time (PT) to fall to 2.0–3.0 INR (international normalized ratio). This may take a few days. ^a
2. Obtain the baseline PT.
3. (a) If the PT is less than 3.1 INR, proceed with surgery and skip to step 6. (b) If the PT is more than 3.0 INR, go to step 4.
4. Stop warfarin approximately 2 days before surgery.
5. Check the PT daily, and proceed with surgery on the day when the PT falls to 3.0 INR.
6. Take extra measures during and after surgery to help promote clot formation and retention.
7. Restart warfarin on the day of surgery.

Patients Receiving Heparin

1. Consult the patient's physician to determine the safety of stopping heparin for the perioperative period.
2. Defer surgery until at least 6h after the heparin is stopped or reverse heparin with protamine.
3. Restart heparin once a good clot has formed.

Box 1.24

Management of Patient With a Seizure Disorder

1. Defer surgery until the seizures are well controlled.
2. Consider having serum levels of antiseizure medications measured if patient compliance is questionable.
3. Use an anxiety-reduction protocol.
4. Take measures to avoid hypoglycemia and fatigue in the patient.

Box 1.25

Management of Patient Who Is Pregnant

1. Defer elective surgery until after delivery, if possible.
2. Consult the patient's obstetrician if surgery cannot be delayed.
3. Avoid dental radiographs unless information about tooth roots or bone is necessary for proper dental care. If radiographs must be taken, use proper lead shielding.
4. Avoid the use of drugs with teratogenic potential. Use local anesthetics when anesthesia is necessary.
5. Use at least 50% oxygen if nitrous oxide sedation is used, but avoid use during the first trimester.
6. Avoid keeping the patient in the supine position for long periods to prevent vena caval compression.
7. Allow the patient to take trips to the restroom as often as needed.

Box 1.26

Dental Medications to Avoid in Patients Who Are Pregnant

Aspirin and Other Nonsteroidal Antiinflammatory Drugs

- Carbamazepine
- Chloral hydrate (if chronically used)
- Chlordiazepoxide
- Corticosteroids
- Diazepam and other benzodiazepines
- Diphenhydramine hydrochloride (if chronically used)
- Morphine
- Nitrous oxide (if exposure is >9h/wk, oxygen concentration is less than 50%, or pregnancy is in the first trimester)

Nonsteroidal Antiinflammatory Agents

- Pentazocine hydrochloride
- Phenobarbital
- Promethazine hydrochloride
- Tetracyclines

Box 1.27

Classification of Medications With Respect to Potential Fetal Risk

Category A: Controlled studies in women have failed to demonstrate a fetal risk in the first trimester (and there is no evidence of risk in later trimesters), and the possibility of fetal harm appears remote.

Category B: Either animal reproduction studies have not demonstrated a fetal risk and there are no controlled studies in pregnant women, or animal reproduction studies have shown an adverse effect (other than decreased fertility) that was not confirmed in controlled studies on women in the first trimester (and there is no evidence of a risk in later trimesters).

Category C: Either studies in animals have revealed adverse fetal effects and there are no controlled studies in human beings, or studies in women and animals are not available. Drugs in this category should only be given if safer alternatives are not available and if the potential benefit justifies the known fetal risk or risks.

Category D: Positive evidence of human fetal risk exists, but benefits for pregnant women may be acceptable despite the risk, as in life-threatening or serious diseases for which safer drugs cannot be used or are ineffective. An appropriate statement must appear in the “warnings” section of the labeling of drugs in this category.

Category X: Either studies in animals or human beings have demonstrated fetal abnormalities, or there is evidence of fetal risk based on human experience (or both); and the risk of using the drug in pregnant women clearly outweighs any possible benefit. The drug is contraindicated in women who are or may become pregnant. An appropriate statement must appear in the “contraindications” section of the labeling of drugs in this category.

From the United States Food and Drug Administration.



Thank you